

ECEN 474/704: Lab 04

Advanced Layout Design Techniques (layout of two stage operational amplifiers)

Sheena Selvakumar

UIN: 537000654

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I. DISCUSSION OF THE LAB AND RESULTS

Guard Rings Purpose:

Used to isolate sensitive analog blocks from substrate noise and prevent latch-up by surrounding PMOS with N-well guard rings (VDD) and NMOS with substrate guard rings (VSS).

Implementation Details:

Guard rings are made using RX_M1 vias, with stacked vias perfectly aligned between layers for proper connection.

Two-Stage Op-Amp Layout:

Layout the given op-amp using good guard ring practices, the provided floor plan, and dummy transistors to ensure symmetry and matching.

Capacitor Design:

Use vncapx capacitor from the cmhv7sf library with M1–MT metal layers to save area, and surround it with guard rings.

Clean DRC Before Additions:

Ensure the amplifier layout is free from DRC errors before adding other structures like capacitors, pads, or ESD circuits.

Pad Addition:

Add VDD and VSS pads first (size 114 μm , type WB), connect using via stacks, and avoid placing pads directly over active circuit areas.

ESD Labeling:

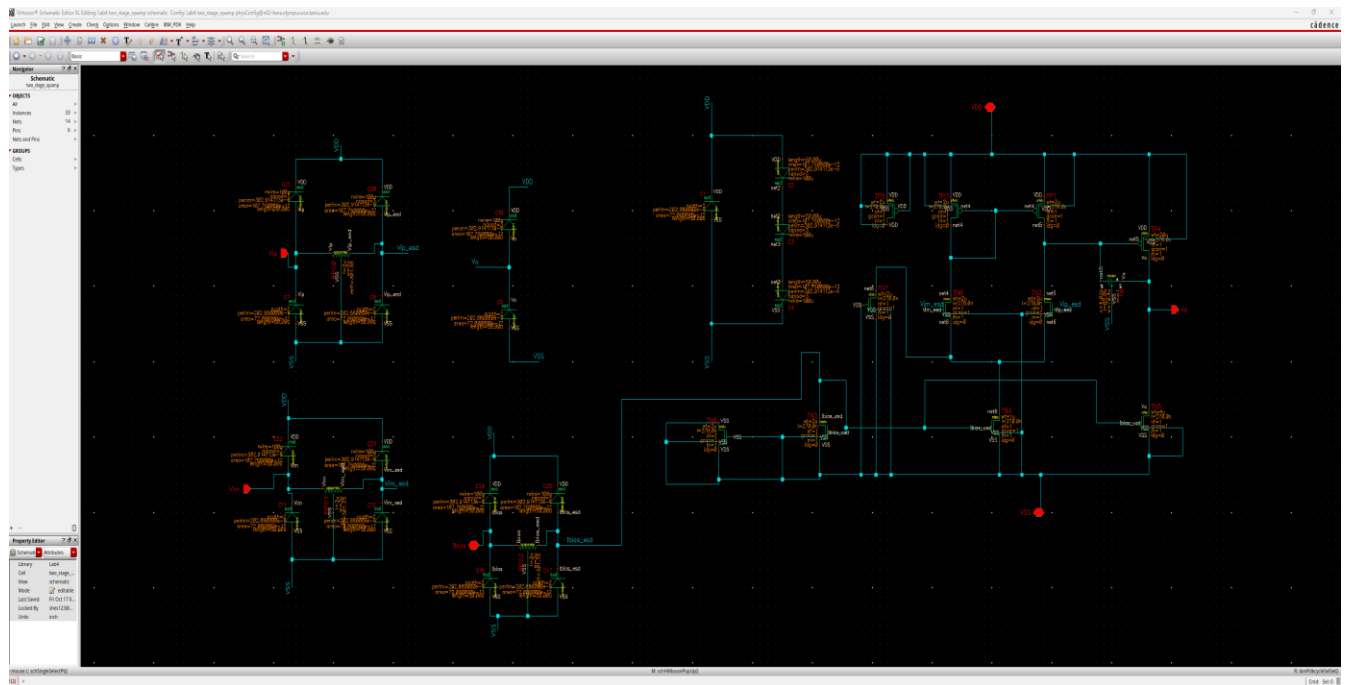
Label power supply pads with “ESD_POWER” on the AM(ESD) layer so DRC can apply ESD protection checks.

ESD Diode Strings:

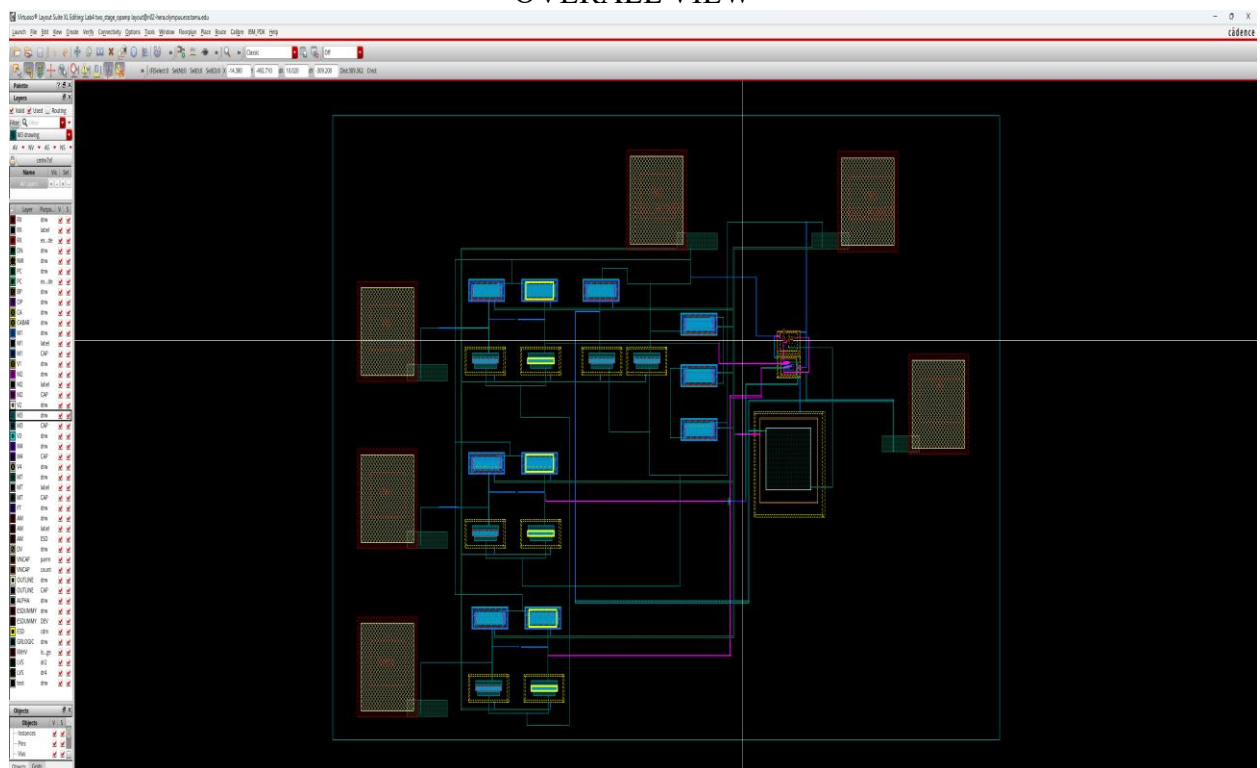
Add esdndsx (VSS→VDD) and esdpdixn (VDD→VSS) diodes to protect against ESD; use 3 diodes in series for the VDD–VSS path.

II. SUMMARY OF RESULTS

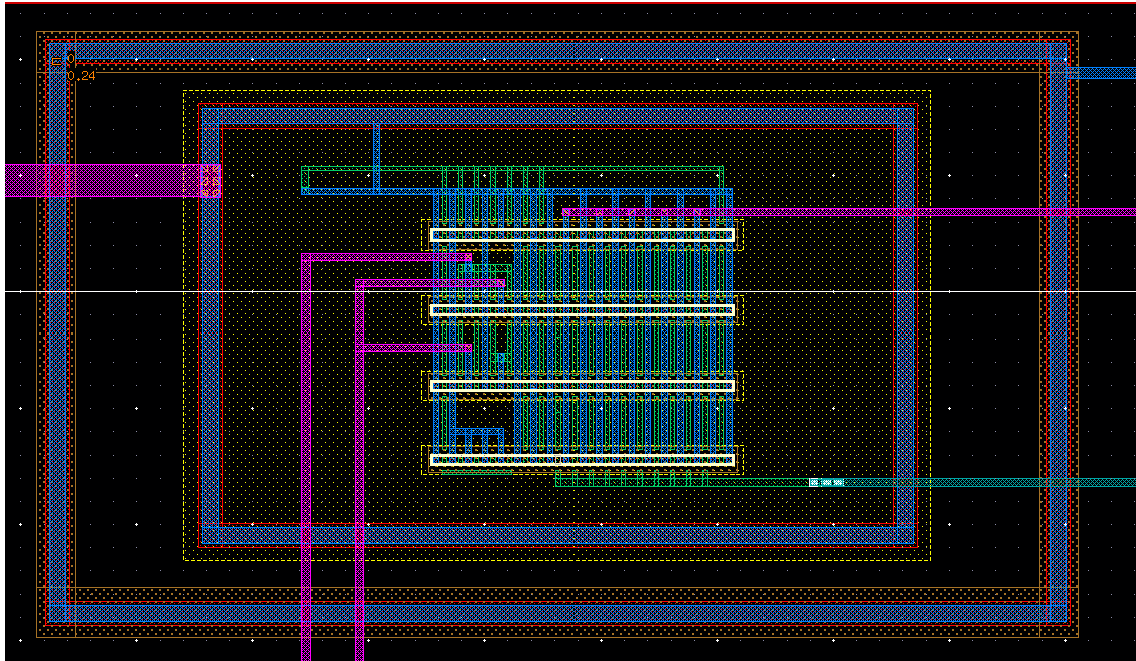
III. SUB-CIRCUIT AND TEST BENCH SCHEMATICS SCHEMATIC OF TWO STAGE OPERATIONAL AMPLIFIERS:



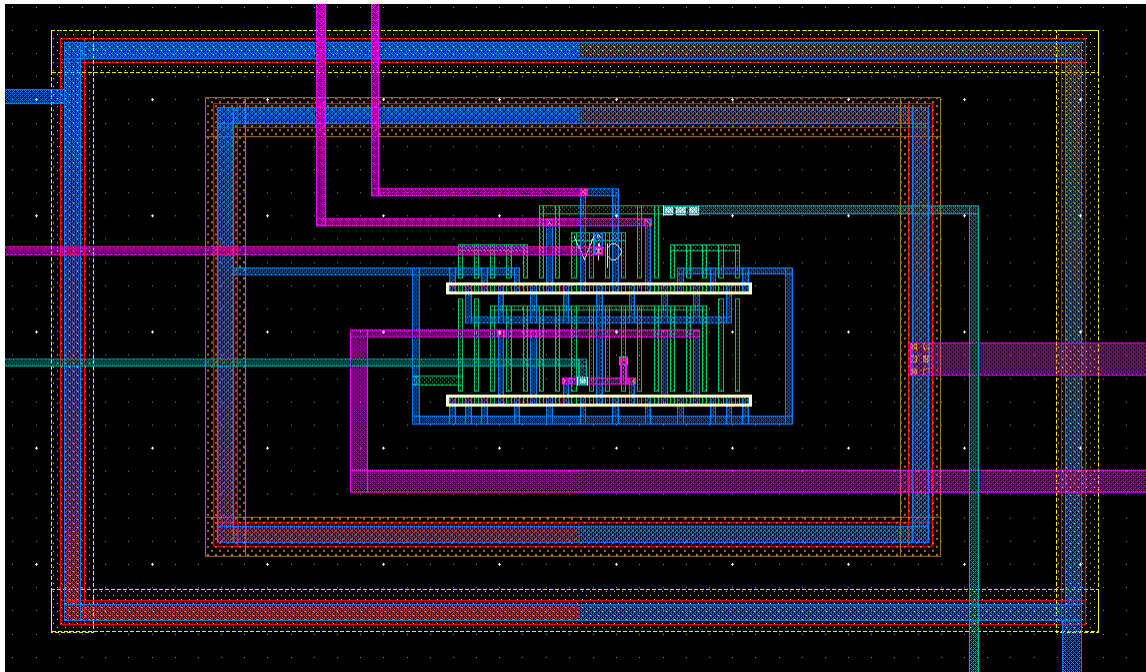
IV. LAYOUT(S) AND FLOOR PLAN(S) OVERALL VIEW



PMOS SECTION CLOSEUP



NMOS SECTION CLOSEUP



V. DRC AND LVS RESULTS

